

# SUDHIR POL

📍 Seattle, Washington    📞 +1 (930) 904-4785    ✉️ sudhirpol522@gmail.com    🔗 linkedin.com/in/sudhir-pol-a6b544179    🌐 sudhirpol522.github.io

## Summary

Machine Learning Engineer with 3 years of experience taking models from prototype to production at Adobe, S&P Global, and American Express. Owns the full path from training and evaluation through containerization, deployment, and monitoring on AWS and Kubernetes, treating latency, cost, and reliability as design constraints.

## Experience

**Adobe** **May 2025 - Aug. 2025**

*Machine Learning Engineer Intern* *San Jose, California*

- Shipped a multi step LangGraph agent over Model Context Protocol and Adobe PDF Spaces that cut an evaluation workflow from 48 hours to 30 minutes, with a LangSmith traced approval gate so stakeholders could reject any plan before it executed
- Built the reliability layer on LiteLLM, putting Gemini 2.5 Pro, GPT 4.1, and Claude 4.5 behind one interface with structured output validation, so failed generations were caught and rerouted instead of reaching users
- Designed an LLM as Judge evaluation harness that gated releases on quality, raising offline F1 score from 79% to 92% by rebuilding the scoring metrics with domain reviewers

**S&P Global** **Nov. 2022 - Aug. 2024**

*Data Scientist II* *Hyderabad, India*

- Owned a Llama 2 extraction service on SageMaker and AWS Lambda end to end, from architecture through deployment and support, removing about 20 minutes of manual lookup per analyst task
- Fine tuned and deployed LayoutLMv3 for token classification on AWS EC2, turning unstructured documents into structured fields that downstream systems could index and query
- Replaced a paid third party vendor with an in house Java extraction service built on named group regex, cutting licensing cost and about 10 minutes of review time per document

**American Express** **Dec. 2021 - Nov. 2022**

*Data Scientist I* *Gurgaon, India*

- Built and containerized the VIBE call quality service with Docker, combining XGBoost with BERT based embeddings, which ran through the year as the system of record for representative bonuses
- Deployed an LSTM model with Flask on AWS EC2 that mapped call transcripts to demand types, letting the business route high priority customer calls first
- Trained and deployed a complaint risk classifier on AWS EC2 with calibrated probabilities, and processed transcript datasets in PySpark to flag missing offer disclosures

## Projects

**MultiDocChat: Production RAG System** | *FastAPI, LangChain, FAISS, Kubernetes, ArgoCD* **May 2025**

- Built and served a RAG system with FastAPI, LangChain LCEL, FAISS indexing, and MMR retrieval, with LangSmith tracing on every request
- Implemented GitOps CI/CD with GitHub Actions, ArgoCD Image Updater, and Kubernetes manifests, achieving zero downtime rolling deployments with semantic versioning

**QuantRoute: Cost Aware Inference Router** | *vLLM, llm-compressor, AWQ, GPTQ, A100* **Nov. 2025**

- Compressed Qwen2.5-7B to 4 bit and served multiple precision lanes on vLLM, reaching 2.2 times faster decode at 56% lower serving cost on financial workloads
- Built a difficulty classifier and escalation loop that sends only hard requests to the expensive lane, using offline oracle analysis to locate routing bottlenecks

## Education

**Indiana University** **Aug. 2024 - May 2026**

*Master of Science in Data Science* *Bloomington, United States*

**Rajarambapu Institute of Technology** **Aug. 2017 - May 2021**

*Bachelor of Technology in Electronics and Tele-Communications* *Sangli, India*

## Writing

- Speculative Decoding: Make LLM Inference Faster *Oct. 2025*
- The Math of AWQ: Protecting Salient Channels from the Inside Out *Jun. 2026*
- Demystifying GPTQ: From Lagrange Multipliers to Vectorized PyTorch *Jun. 2026*

## Skills

**Languages:** Python, C++, SQL, Java, PySpark

**Machine Learning:** Deep Learning, NLP, Transformers, LLMs, RAG, Fine Tuning, Model Optimization, Evaluation and Metric Design

**Frameworks:** PyTorch, TensorFlow, HuggingFace, Scikit-learn, XGBoost, LangChain, LangGraph, LiteLLM, vLLM, FastAPI

**Infrastructure and MLOps:** AWS (SageMaker, EC2, Lambda), Docker, Kubernetes, ArgoCD, GitHub Actions, CI/CD, MLflow